Business Process Automation

with Power Automate and Copilot Studio Agents

WSQ Course Code: TGS-2022017524 · 2-Day Hands-On Training · Modules 1–5

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer

?

Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

AA

Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE
Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION
Microsoft Power Platform, AI and data analytics — 20+ years of training experience.

DELIVERS
WSQ courses on AI, business automation, data science and software engineering.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with Power Automate, Copilot Studio or other automation tools (if any).
- One repetitive business process you would love to automate after this course.

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

COURSE OUTCOMES

What You'll Learn

Automate the workflow → add AI → **Build the agent** → expose it → **Connect it over HTTP** → make it true → **Ground it with RAG**

1 Design

Read a business process and decide which parts must be deterministic and which may be AI.

2 Build

Create Power Automate flows and Copilot Studio agents with tools, skills and knowledge.

3 Govern

Place a human where the consequence lands, and test what the agent must refuse.

By the end you can build the automation, add the agent, and defend where you put the human.

THE JOURNEY

Eleven Connected Labs

0 **Environment Setup**
Developer environment, Dataverse, credits

1 **Trigger and Actions**
Form → email; dynamic content

2 **Log to Excel**
Commit the audit row before confirming

3 **Leave Application Approval**
The flow pauses for a manager

4 **Agents**
Instructions · skills · knowledge · tools

5 **Invoke Agents**
SharePoint grounding, published to Teams

6 **HTTP + Approval Agent**
Structured output; the AI never writes the record

7 **HTTP + Chatbot**
The agent alone, in public

8 **HTTP + Human Review**
The run that will not proceed

9 **RAG with Knowledge Base**
Three nodes, no ingestion

10 **RAG with Pinecone**
Chunking, embeddings, top_k

Eleven labs, each reusing the one before it. Nothing is rebuilt from scratch.

How This Course Works

1 Concept
The idea and the decision rule, before the product is opened.

2 Pattern
The same idea drawn as a business workflow you could hand to a colleague.

3 Lab
You build it, in the numbered lab named on the concept slide.

4 Probe
You try to break it — the refusals and wrong answers are the lesson.

5 Reuse
The working artefact becomes the starting point of the next lab.

Concepts first. Every lab in this course is an application of a concept you have already seen.

The probes matter as much as the builds. An agent that answers ten questions correctly and the eleventh confidently wrong has not been tested — it has been demonstrated.

SCHEDULE

Lesson Plan — 2 Days, 9:00am–6:00pm

Day 1 — Automate, then add the agent

9:00 – 9:45	Welcome, admin, course map
9:45 – 10:40	Module 1: BPA, Power Platform, flows, triggers, actions
10:55 – 12:15	Lab 0 · Lab 1 — environment, trigger and actions
12:15 – 12:45	Module 1: dynamic content, run history, commit order
1:45 – 3:15	Lab 2 · Module 2 — Excel logging, conditions, human in the loop
3:30 – 6:00	Lab 3 · Module 3 · Lab 4 — approval, agent anatomy, four agents

Day 2 — Connect, supervise, ground

9:00 – 9:40	Recap · Lab 5 — grounding and publishing to Teams
9:40 – 11:10	Module 4 · Lab 6 — HTTP, structured output, boundary of agency
11:25 – 12:05	Lab 7 — the agent alone in public
1:05 – 2:05	Lab 8 — the human review gate
2:05 – 3:55	Module 5 · Lab 9 · Lab 10 — RAG, built-in then Pinecone
4:00 – 6:00	Written Assessment (SAQ) 1 hr + Practical Performance 1 hr

Each day runs 9:00am–6:00pm — 8 scheduled hours after a 1-hour lunch, with the two 15-minute tea breaks counted within.

Access the Hands-On Labs

1 **11 labs**

Lab 0 setup plus Labs 1–10, each with its own step-by-step guide.

3 **Test data**

Sample question sets and probe cases written to make each agent fail.

2 **Ready assets**

Excel workbooks, brochure PDFs, HTML pages, JSON schemas and flow packages.

4 **Work in order**

Open labs/README.md and follow the Next link at the foot of each lab.

Everything is in the course repository; the Learner Guide reproduces every lab in full.

Your Training Account

One assigned Microsoft 365 training account per learner — the same login for every lab.

training1@tertiaryinfotech.onmicrosoft.com	Reserved for the trainer	**training2@tertiaryinfotech.onmicrosoft.com**	Learner account 1
training3@tertiaryinfotech.onmicrosoft.com	Learner account 2	**training4@tertiaryinfotech.onmicrosoft.com**	Learner account 3
training5@tertiaryinfotech.onmicrosoft.com	Learner account 4	**training6@tertiaryinfotech.onmicrosoft.com**	Learner account 5
training7@tertiaryinfotech.onmicrosoft.com	Learner account 6	**training8@tertiaryinfotech.onmicrosoft.com**	Learner account 7
training9@tertiaryinfotech.onmicrosoft.com	Learner account 8	**training10@tertiaryinfotech.onmicrosoft.com**	Learner account 9

Password for training2 – training10: Tertiary@0808

Sign in at copilotstudio.microsoft.com and make.powerautomate.com with the account the trainer assigns you — stay in the shared training environment and do not change the password.

MODULE 1

Business Process Automation and Power Automate

What automation actually removes · the Power Platform · flows, triggers and actions · dynamic content and run history

Applied in Lab 0, Lab 1 and Lab 2

What Business Process Automation Removes

Business process automation is not “software that does work faster”. It is the removal of the hand-off where a person re-types what another system already knows.

MANUAL

Form filled in → Someone reads it → Re-typed into Excel → Someone remembers to reply

AUTOMATED

Form submitted → Flow triggered → Row written → Reply sent

Consistency

The same input produces the same output, every time, at 3am.

Traceability

Every run leaves a history someone can read a year later.

Capacity

People are moved to the judgement calls the machine cannot make.

MODULE 1 · CONCEPT

The Power Platform, and the Environment That Holds It

1 Power Automate

Workflows. Triggers and actions that run without a person.

Labs 1–3

2 Copilot Studio

Conversational agents with instructions, knowledge and tools.

Labs 4–10

3 Dataverse

The managed data store behind the environment.

Lab 0

4 Connectors

Outlook, Excel, SharePoint, Teams, Forms, HTTP — 1,000+.

Every lab

THE ENVIRONMENT IS THE CONTAINER

Flows, agents and data all live inside one environment. Power Automate and Copilot Studio must be pointed at the same one, or your agent cannot see your flow. This is the single most common "where did it go?" in the course.

LAB 0

Lab 0 — Environment Setup

Copilot Studio Training (Developer) environment · Power Platform admin centre · 40 min

Every Flow Is the Same Four Parts

outputs outputs outputs

TRIGGER
what starts it

ACTION
do the work

ACTION
do more work

OUTPUT
notify or return

Every flow, in every lab, is this shape. Only the trigger and the actions change.

1 One trigger

A flow has exactly one trigger. Change the trigger and you have a different flow — the actions may not change at all.

2 Actions are ordered

Each action runs after the one above it, and can read the outputs of every step before it.

3 Dynamic content

Those earlier outputs are inserted as tokens, not typed. A typed value is a constant that will be wrong tomorrow.

Trigger, actions, outputs. Learn this once in Lab 1 and every later lab is a variation.

Triggers — What Is Allowed to Start a Process

Four trigger families. The one you choose is a statement about who or what is allowed to start the process.

Manual

A person presses Run

Instant cloud flow · button in the mobile app or Teams

Scheduled

A clock reaches a time

Recurrence — set the time zone or it runs on UTC

Automated

An event happens in a system

New form response · new email · new SharePoint item · new file

Request

Something calls in from outside

HTTP request received · When an agent calls the workflow

Trigger used in this course	Where	What it means
Automated — new form response	Labs 1, 2, 3	A business event starts the process
Request — HTTP request received	Labs 6, 7, 8, 10	A website posts JSON and waits for JSON back
Request — when an agent calls the workflow	Labs 4, 6	A conversation decides to call a tool
Scheduled / Manual	Reference	A clock, or a person, starts it deliberately

Actions — The Six Families You Will Use

An action either moves data, decides something, waits for a person, or calls something outside the flow.

Data actions

Compose · Parse JSON · Initialize variable · Select · Filter array

Shape and normalise values before anything trusts them.

Connector actions

Send an email (Outlook) · Add a row (Excel) · Create item (SharePoint)

Do the real work in a business system.

Control actions

Condition · Switch · Apply to each · Scope · Terminate

Decide which path the run takes.

Human actions

Start and wait for an approval · Human review (agent flows)

Suspend the run until a person responds.

Integration actions

HTTP · Response · Invoke another flow · Run a prompt (AI Builder)

Reach outside the Power Platform, or answer the caller.

Agent actions

Agent node · Respond to the agent · structured output

Let the model decide, inside a flow that does not.

You will use every one of these families by the end of Lab 10.

Dynamic Content — The Token, Not the Text

Trigger runs → (produces) → Outputs exist → (picker) → Later action references them → (resolved) → Value arrives at run time

✓ INSERTED WITH THE PICKER

The value renders as a coloured token. It resolves at run time to whatever the earlier step actually produced.

Every dynamic value in every lab is inserted this way.

✗ TYPED BY HAND

It stays dead text. The flow runs green and the value arrives empty — because a reference to nothing resolves to empty, not to an error.

This costs more class time than any other single mistake.

A green run is not a correct run. Open the run history and read what each step actually received.

LAB 1

Lab 1 — Trigger and Actions

Form to email confirmation · Power Automate + Microsoft Forms + Outlook · 40 min

Run History — The Only Honest Account

The run history is the only honest account of what happened. Every lab in this course is verified from it.

1 Open the run

My flows → the flow → Run history, or the Activity tab in an agent flow.

2 Read each step

Expand a step to see its inputs and its outputs — what it was given, what it produced.

3 Find the empty one

A wrong answer usually traces to a step whose input was blank, not to a red error.

4 Fix the reference

Re-insert the token with the picker, republish, and run one test — one change per cycle.

THE THREE STATES THAT LOOK ALIKE

Succeeded with the right data · Succeeded with empty data · Still Running because it is waiting for a person.
Only the first is done. The second is the dangerous one — it looks exactly like success.

LAB 2

Lab 2 — Log to Excel

Log the enquiry, then send the email · Power Automate + Excel Online · 45 min

Order Matters — Commit Before You Confirm

The order of two actions is a business decision, not a technical one.

LOG, THEN CONFIRM

Write the row → **Send the email**

If the email fails, the enquiry is still on the register and someone can chase it. You have a record of a missed reply.

CONFIRM, THEN LOG

Send the email → **Write the row**

If the row fails, you have promised a customer a reply that nobody can see. The evidence of the promise is gone.

Commit the record of the obligation before you create the obligation. This is Lab 2 in one sentence.

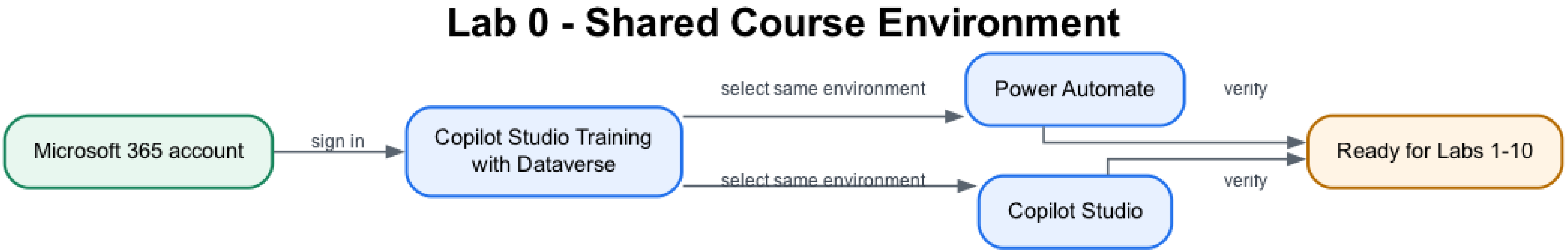
LAB 2

Lab 2 — Log to Excel

Log the enquiry, then send the email · Power Automate + Excel Online · 45 min

Lab 0 — The Shared Course Environment

One Copilot Studio Training (Developer) environment, with Dataverse and Copilot Credits, stands behind every later lab.



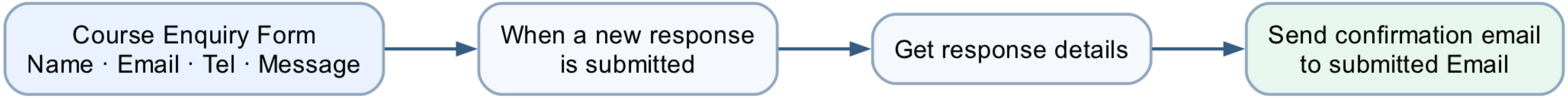
LAB 0

Lab 0 — Environment Setup

Copilot Studio Training (Developer) environment · Power Platform admin centre · 40 min

Lab 1 — Form to Email Confirmation

A trigger starts a flow; actions consume the trigger's outputs as dynamic content.



LAB 1

Lab 1 — Trigger and Actions

Form to email confirmation · Power Automate + Microsoft Forms + Outlook · 40 min

Lab 2 — Log the Enquiry, Then Send

Commit before you confirm — the audit row is written before the acknowledgement is sent.

LAB 2

Lab 2 — Log to Excel

Log the enquiry, then send the email · Power Automate + Excel Online · 45 min

MODULE 2

Control Flow and Human in the Loop

Conditions and branching · approvals · in / on / out of the loop · where the human belongs

Applied in Lab 3

Conditions — Both Paths Must Exist

A condition splits one run into two paths. Both paths must be built — including the one you hope never runs.

Condition
is the value X?

IF YES

The approved path. Continue the business process.

IF NO

The rejected path. Notify, record and stop — never fall silent.

Control action	What it does	Example in this course
Condition	One test, two branches	Approved or rejected
Switch	One value, many branches	Leave type: Annual / Medical / Unpaid
Apply to each	Repeat actions over a list	Every row returned by a lookup
Terminate	End the run deliberately	Stop with a status a reader can interpret

Human In, On, and Out of the Loop

Three arrangements that people use interchangeably, and that are not the same thing.

Pattern	Who decides	Who acts	Can AI proceed alone?
Human IN the loop	AI proposes, human decides	Human authorises, system acts	**NO — it blocks**
Human ON the loop	AI decides and acts	Human monitors, can intervene	**Yes — after the fact**
Human OUT of the loop	AI decides and acts	AI acts	**Yes — nobody checks**

WHERE THIS COURSE PUTS THE HUMAN

Lab 3 — a manager approves leave. Lab 8 — a licensed adviser approves a draft before it is sent.
Labs 6, 7, 9 and 10 are deliberately out of the loop, so you can see what that costs.

LAB 3

Lab 3 — Leave Application Approval

The flow pauses for a manager · Power Automate + Approvals + Outlook · 45 min

How an Approval Suspends a Running Flow

action	pause	decision	resume	
Request submitted	**Start and wait for an approval**	**The run SUSPENDS**	**A person responds**	**Condition reads the Outcome**

OUTCOME = APPROVE

Send the approval message, update the record and continue the business process.

OUTCOME = REJECT

Send the rejection with the approver's comments, and route it to a named person — never to silence.

SEND APPROVALS TO TEAMS, NOT OUTLOOK

Verified on a live tenant: Outlook created the request and never delivered the mail — the run sat at Running, looking healthy.
Every human gate in these labs uses the Microsoft Teams Approvals app.

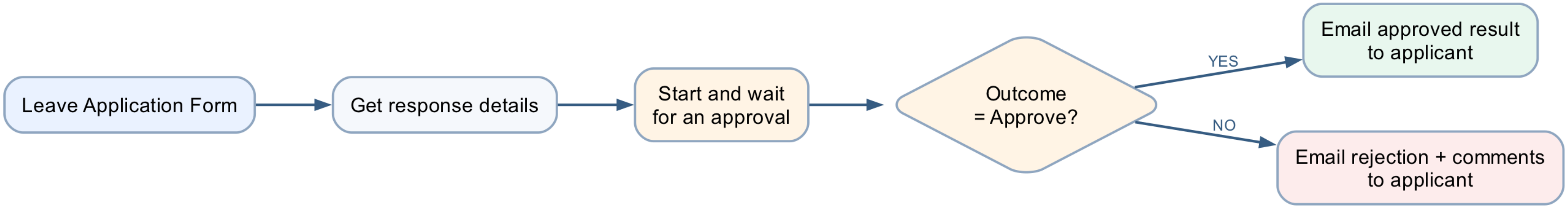
LAB 3

Lab 3 — Leave Application Approval

The flow pauses for a manager · Power Automate + Approvals + Outlook · 45 min

Lab 3 — Leave Application Approval

A running flow suspends at the approval action and branches on the human decision.



LAB 3

Lab 3 — Leave Application Approval

The flow pauses for a manager · Power Automate + Approvals + Outlook · 45 min

MODULE 3

Copilot Studio Agents

The new Copilot Studio — model, harness, designer · workflow versus agent · instructions, skills, knowledge, tools · what is enforced · publishing

Applied in Lab 4 and Lab 5

Workflow or Agent — They Fail Differently

WORKFLOW — Power Automate

- You decide the path in advance
- The same input always gives the same output
- It can only do what you built
- It fails loudly, at a named step
- You test it by checking the result

AGENT — Copilot Studio

- It chooses the path at run time
- The same input may give a different answer
- It can combine what it was given in new ways
- It fails quietly, with a confident wrong answer
- You test it by trying to break it

Use a workflow where the rule is known. Use an agent where the language varies. Most real systems need both.

Labs 1–3 are pure workflow. Labs 4–5 are pure agent. Labs 6–10 are both, and the seam between them is the lesson.

The New Copilot Studio — a Model Inside a Harness

Copilot Studio now separates the part that reasons from the part that governs. Two ideas name that split.

1 The AI model — the brain

The reasoning engine. It understands language, spots patterns, generates content and decides what to do next. You choose it per agent — GPT-5.x or Claude — from the Model dropdown.

2 The harness — around it

The orchestration layer the model works inside: context, tools, governance, workflows and security. The harness is model-agnostic — it turns reasoning into governed work.

ONE MODEL, THREE HARNESSES IT CAN WORK INSIDE

GitHub Copilot harness

Reasoning-heavy agents and workflows. Takes a goal, breaks it into steps, uses skills and memory, edits files in a governed sandbox. Billed in Copilot Credits.

Standard harness

Rule-based agents and structured, repeatable conversations. You author the topics and paths, so the behaviour is predictable for well-understood requests.

Copilot chat harness

Extends Microsoft 365 Copilot Chat with your enterprise knowledge, so employees get grounded answers inside the chat they already use.

Everything you build in Labs 4–10 is a model you chose, inside a harness you configure.

Choosing a Harness

Consideration	GitHub Copilot harness	Standard harness	Copilot chat harness
Best for	Complex, multi-step business processes	Rule-based agents, structured conversations	Extending M365 Copilot Chat
How it works	Reasons through a goal, step by step	Follows the topics and rules you define	Connects enterprise knowledge to chat
When a step fails	Retries, or finds another path	Follows only the paths you built	Not a focus
Works with files	Creates and edits Word, Excel, PowerPoint, PDF	Not a focus	Not a focus
Skills and memory	Yes — core building blocks	Not a focus	Not a focus
Billing	Copilot Credits — you pay for usage when you build and when you publish	Copilot Studio licensing	Consumption, or M365 Copilot seats

Start with the simplest harness that meets the need — a retrieval agent that only answers from documents does not need the reasoning loop.

Modern Orchestration — the Loop Replaces the Plan

The standard harness builds a plan, then runs it. The GitHub Copilot harness keeps deciding from the latest state.

STANDARD — PLAN, THEN EXECUTE

1. INTENT
understand

2. PLAN
synthesise steps

3. EXECUTE
run the plan

An explicit plan object: collect inputs, execute actions, summarise. Less live replanning and easier to inspect — but a detour or a failed step ends the plan.

GITHUB COPILOT — THOUGHT · ACTION · OBSERVATION

THOUGHT → ACTION → OBSERVE → DECIDE → THOUGHT

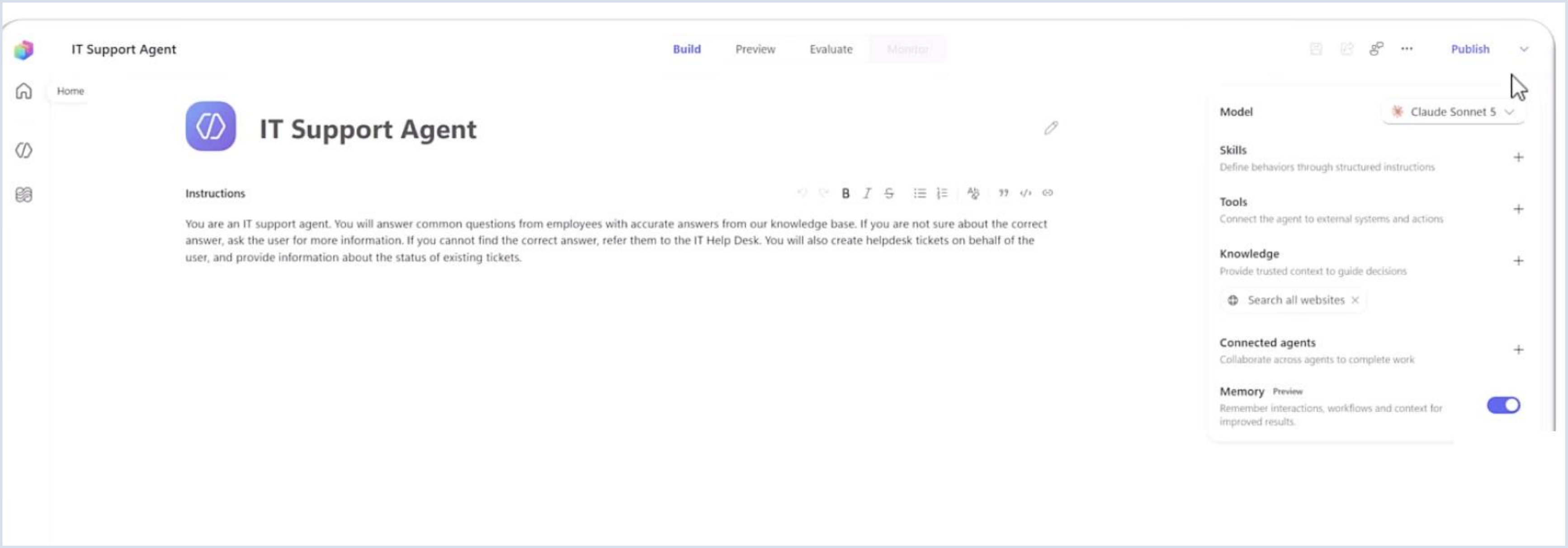
latest state

Reason over the latest state, call a tool, read the result, choose the next step live. It retries or reroutes on failure, and picks the task back up after a detour.

What the loop buys: fewer, smarter questions (it infers what it already knows) · handles detours without losing the task · chains and parallelises tools · recovers from errors instead of surfacing them.

The loop fits problem spaces you cannot map upfront. Where the path is known, that freedom is cost, not value — keep it in a workflow.

The New Agent Designer



One page

Skills, Tools, Knowledge, Connected agents and Memory, all on one rail.

Four tabs

Build · Preview · Evaluate · Monitor — author, probe, score, then watch it live.

Model picker

GPT-5.x or Claude, chosen per agent — the harness stays the same.

Memory (preview)

Per-user remembered context. A maker toggle — off means unused.

Everything Lab 4 builds — instructions, skills, knowledge, tools — has a single home in this designer.

The Anatomy of an Agent

INSTRUCTIONS

Who it is and what it must
never do. Always in force.

SKILLS

Named procedures, applied
when the topic matches.

THE AGENT
name · model · description

KNOWLEDGE

Documents it may read.
It cannot read anything else.

TOOLS

Flows it can call to act
outside the conversation.

A fifth part — CONNECTED AGENTS — lets one agent hand a conversation to another with its own knowledge and audience.

LAB 4

Lab 4 — Agents

Procurement, HR, Sales and IT Support · Copilot Studio agents + agent flows + SharePoint + Teams · 45 min

Which Parts of an Agent Hold — and Which Do Not

The spine of Lab 4: which of an agent's five parts can the model ignore?

Part	What it is	Enforced?
Instructions	Who the agent is, always in force	**NO — probabilistic**
Skill	A named procedure for a matching topic	**NO — the model decides it applies**
Knowledge	Documents the agent may read	**PARTLY — it cannot read what it lacks**
Tool	A flow that acts outside the conversation	**YES — the flow's own logic is enforced**
Connected agent	A separate agent, its own knowledge	**YES — the knowledge boundary is real**

A control the model cannot reach beats a rule you asked it to follow.

So: a tool the agent does NOT have is a control. And the schema beats the prompt — a field that exists will eventually be filled.

Writing Instructions That Constrain

Instructions are prose, but they are not free text. Four things every agent instruction in this course states.

1 Identity

"You are an HR policy information assistant."
One role, stated in one line.

2 Source rule

"Answer using only the approved SharePoint HR policy source."
Where facts may come from.

3 Refusals

"Do not expose, request or infer personal employee records."
Stated as prohibitions, not preferences.

4 Escalation

"When the source is insufficient, say so and direct the user to HR."
Where the conversation ends when it cannot continue.

NEVER PASTE @{...} INTO AN INSTRUCTIONS BOX

It is a rich-text editor: it escapes underscores in node names, and a reference to a node that does not exist resolves to EMPTY rather than erroring. Build the prose with gaps and insert every value with the ⚡ picker.

MODULE 3 · CONCEPT

Skills — Instructions on Demand

TOPICS

authored conversational paths — the legacy way to build chatbots

SKILLS

reusable procedures, loaded when the scenario matches

Topics were built for scripted chatbots — not for how these models think. A skill gives the agent a specific, repeatable task and the material to carry it out.

What a skill is

Name + description are the routing metadata; the instructions are the core, with examples, resources and scripts alongside. Metadata sits in context — the full content loads only when the scenario fires.

Why it works

The instruction page stays short and readable, and context loads on demand. A large always-on prompt makes the model weigh irrelevant guidance on every turn.

What to avoid

Vague descriptions (the skill mis-fires or never triggers) · mega-skills that do everything · fact dumps — a skill guides, it does not replace a knowledge source.

A skill is still not enforced — the model decides that it applies. The enforcement table you just saw is unchanged.

Knowledge — Giving Facts and Removing Sources

Documents in SharePoint → (point at it) → Attached as a knowledge source → (wait for Ready) → Indexed by the product → (at run time) → Retrieved when a question matches

1 It is a boundary

The agent genuinely cannot read a document you did not give it. This is the one part of the agent that is close to enforced.

2 Turn off general knowledge

Otherwise the model answers from what it learned in training, and you cannot tell which answers those were.

3 Remove "Search all websites"

It is on by default. A fee from the open web is indistinguishable from a fee in your own brochure.

Grounding is not only about giving the agent facts. It is about taking away every other source of them.

LAB 5

Lab 5 — Invoke Agents

HR Support Agent grounded in SharePoint · Copilot Studio + SharePoint + Teams · 30 min

Tools — Where the Agent Stops and the Flow Starts

User asks for something → *intent* → **Agent decides a tool applies** → *typed inputs* → **Flow runs — deterministic** → *Respond to the agent* → **Result returns to the conversation**

THE AGENT'S PART

Decides that a tool is relevant, and fills its inputs from what the person said.

This part is probabilistic. It may call the wrong tool, or call the right one with the wrong values.

THE FLOW'S PART

Validates, looks up, applies the threshold, writes the audit row, sends to the approver.

This part is enforced. Whatever the agent believed, the flow's own logic still runs.

The tool description is how the agent knows when to call it — write it for the model, not for a developer.

LAB 4

Lab 4 — Agents

Procurement, HR, Sales and IT Support · Copilot Studio agents + agent flows + SharePoint + Teams · 45 min

Multiple Agents — A Split Is a Governance Decision

One agent that knows everything has no boundaries. Splitting is how you give it some.

HR AGENT
the parent

- **Policy & Benefits**
- **Leave**
- **Onboarding**
- **Payroll**

Knowledge is separated

Each child reads only its own documents. That boundary is real.

Conversation is not

What the employee said still flows across. Privacy is not automatic.

A split is governance

You are deciding who may know what — not tidying a big prompt.

LAB 4

Lab 4 — Agents

Procurement, HR, Sales and IT Support · Copilot Studio agents + agent flows + SharePoint + Teams · 45 min

Publishing to Teams

Build and save → probe it → **Test in Preview** → version → **Publish** → Teams / web → **Add a channel** → verify → **Users reach it in Teams**

1 Preview is not published

Changes are invisible to Teams users until you publish again. A stale answer in Teams usually means an unpublished edit.

2 One change per cycle

Each publish-and-test cycle costs a publish plus about 25 seconds. Two simultaneous edits make a failure uninterpretable.

3 Test as the user

Open it from Teams with the account a real user would have — not from the maker's Preview pane.

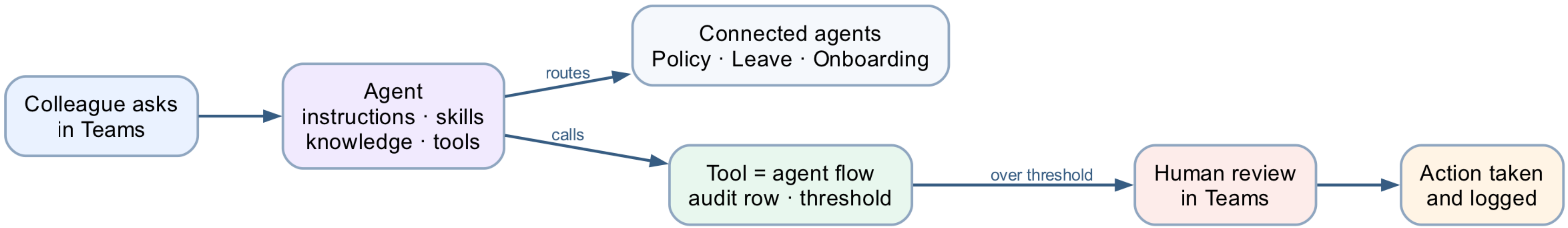
LAB 5

Lab 5 — Invoke Agents

HR Support Agent grounded in SharePoint · Copilot Studio + SharePoint + Teams · 30 min

Lab 4 — Four Agents, One Spine

An agent is instructions, skills, knowledge, tools and connected agents — and only some of those are enforced.



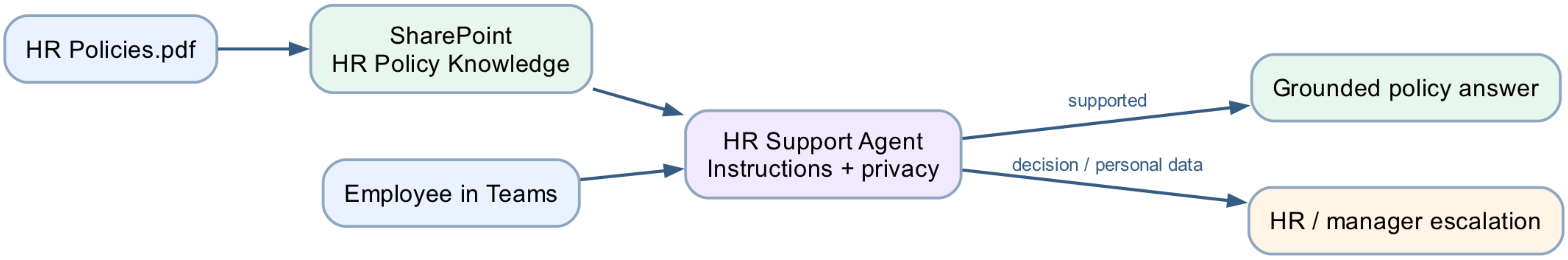
LAB 4

Lab 4 — Agents

Procurement, HR, Sales and IT Support · Copilot Studio agents + agent flows + SharePoint + Teams · 45 min

Lab 5 — Grounding and Publishing an Agent

Ground an agent in a permission-controlled SharePoint source, test its refusals, and publish it to Teams.



LAB 5

Lab 5 — Invoke Agents

HR Support Agent grounded in SharePoint · Copilot Studio + SharePoint + Teams · 30 min

MODULE 4

Agent Flows, HTTP and the Boundary of Agency

The new workflow designer · the Agent node inside a flow · HTTP request and response · structured output · the human review gate

Applied in Lab 6, Lab 7 and Lab 8

Agent Flows — The Model Inside the Workflow

An agent flow is a Power Automate flow that may contain an Agent node — the model, inside the workflow.

JSON in | clean | facts | structured | act

HTTP trigger → **Compose normalise** → **SharePoint lookup** → **AGENT apply the rules** → **Condition on the decision** → **Response**

WHAT THE AGENT NODE IS FOR

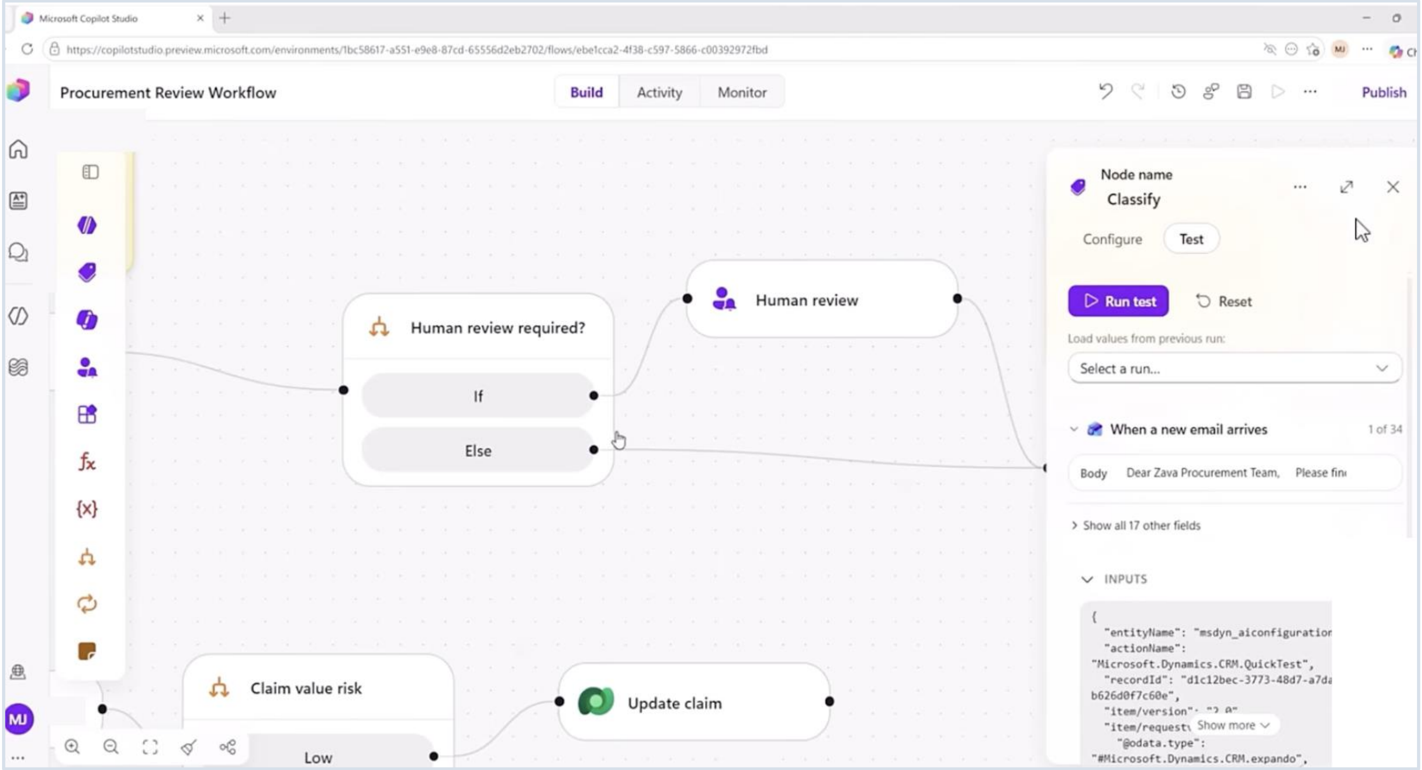
Judgement over language: reading a free-text reason, applying ordered rules, classifying a tone, drafting a reply. Anything where the input varies more than a form allows.

THE COPILOT CREDITS TRAP

Agent flows consume Copilot Credits on every run. Many Default environments have none and fail with InsufficientMcsCredits. A licence does not fix it — licences are per-user, credits are per-environment.

This is why Lab 0 builds a Copilot Studio Training (Developer) environment rather than a Sandbox.

The New Workflow Designer



A visual canvas

Deterministic steps drawn as nodes — AI only where a step needs judgement.

Node-by-node testing

Run test on one node, with values loaded from a previous run.

Agents inside workflows

Drop an inline agent into one step, or call a published agent as a tool.

MCP-enabled tools

Connectors and MCP tools, inside the Microsoft security boundary.

Workflows keep the structure of agent flows and add the intelligence of agents — without the unpredictability.

MODULE 4 · CONCEPT

HTTP — A Website Calls Your Flow and Waits

Website form → (submit) → POST JSON → (over HTTPS) → HTTP trigger receives it → Flow runs → (build reply) → Response returns JSON → (render) → Page updates

Part	Meaning	In these labs
Request	What the caller sends	A JSON body matching the schema you declared
Schema	The shape you promise to accept	Generated from a sample payload — then frozen
Response	What the caller gets back	A JSON body the page can read, plus a status code
Synchronous	The page waits	If nothing responds, the browser sees a timeout, not a result

TWO FAILURES THAT LOOK THE SAME FROM THE BROWSER

"Failed to fetch" WITH a successful run = CORS; serve the page from SharePoint, not localhost. "Failed to fetch" with NO run at all = the flow is saved but not published, or the URL is wrong.

Structured Output — Fields, Not Prose

Structured output turns the model's answer from prose into named fields the rest of the flow can branch on.

PROSE — unusable downstream

"I think this application should probably be approved, although the address looks a little unusual and you may wish to check it."

What does a Condition test against that?

STRUCTURED — branchable

applicationId : APP-10432
decision : REVIEW
reason : address unverified
riskFlags : [ADDRESS_MISMATCH]

A Condition can read decision. A person can audit reason.

FOUR DECISIONS, NOT TWO

APPROVED · REJECTED · DUPLICATE · REVIEW
A politically exposed person is not a rejection — it is a case for a human. An agent given only two outcomes will force every ambiguous case into one of them, and you will never see the ones it got wrong.

LAB 6

Lab 6 — HTTP and Application Approval Agent

Marina Trust Bank customer onboarding · Copilot Studio agent flow + SharePoint + Outlook · 60 min

The Boundary of Agency

The most important design decision in the course: what the model is allowed to determine, and what it is not.

THE AI DECIDES — what to do

- Which of six ordered rules applies
- Whether the case needs a human
- How to phrase the reason
- What risk flags to raise

THE FLOW DOES — what must always happen

- Normalise the identifier (Compose)
- Check the register for a duplicate
- Write the customer record
- Write the audit row, before the gate

THE RECORD IS WRITTEN FROM THE COMPOSE ACTION, NEVER FROM THE MODEL'S ANSWER

So an invented identifier has no route into the customer master. The agent's opinion reaches the decision field; it never reaches the data. If you remember one sentence from Day 2, make it this one.

LAB 6

Lab 6 — HTTP and Application Approval Agent

Marina Trust Bank customer onboarding · Copilot Studio agent flow + SharePoint + Outlook · 60 min

The Agent Alone, in Public

Lab 7 is the agent working in public with nobody reviewing it. Four nodes, and two non-negotiable rules.

Chat widget → POST → HTTP trigger → Compose session → instruction · knowledge · history → AGENT rules + facts → reply → Response

Chat widget › **HTTP trigger** › **Compose session** › **AGENT rules + facts** › **Response**

POST

instruction · knowledge · history

reply

RULES LIVE IN THE INSTRUCTION

Collect name, phone and email before answering. Never recommend a product, predict a return or allocate savings. A refusal that must be retrieved can miss.

FACTS LIVE IN THE KNOWLEDGE SOURCE

Fees, timelines and services come from the FAQ PDF. Change the PDF and the answers change — without touching the instruction.

The test cases in Lab 7 are written to make it fail. Read the failures aloud — that is the debrief.

LAB 7

Lab 7 — HTTP and Chatbot

Investment advisor chatbot · Copilot Studio + SharePoint knowledge + static website · 40 min

The Human Review Node — A Gate That Blocks

Lab 8 adds the person back. The Human review node does nothing at all except wait.

THE AGENT'S TERRITORY

classify | read tone | flag | DRAFT

THE GATE

Human review approve / reject in Teams

THE CONSEQUENCE

send · log · hand to a named person

THE TEST OF A REAL GATE

Submit an enquiry, then open Activity. The run says Running — and it will still say Running tomorrow. Nothing times out, nothing defaults, nothing proceeds. A workflow that has done all of its work and will not take the last step.

That pause is the deliverable. Rejected drafts go to a NAMED person who must phone the client — never to silence.

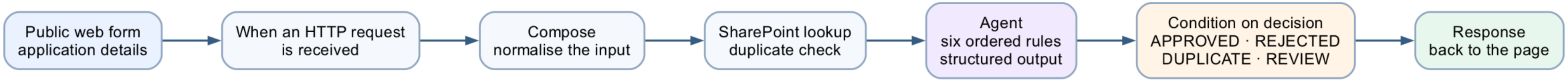
LAB 8

Lab 8 — HTTP and Human Review

Client rapport assistant with a Teams approval gate · Copilot Studio + Human review + Excel Online + Outlook + Teams · 60 min

Lab 6 — Application Approval Agent

The boundary of agency — the AI decides what to do; deterministic actions do what must always happen.



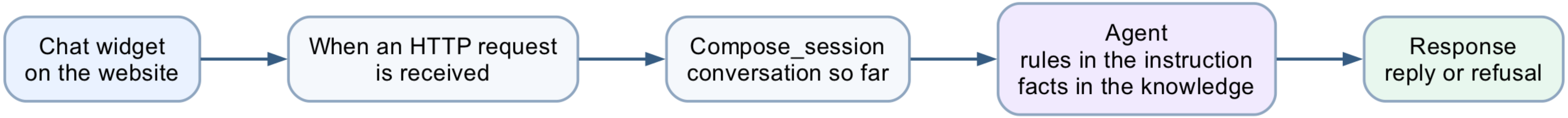
LAB 6

Lab 6 — HTTP and Application Approval Agent

Marina Trust Bank customer onboarding · Copilot Studio agent flow + SharePoint + Outlook · 60 min

Lab 7 — The Chatbot, Working Alone

The agent alone — rules in the instruction, facts in the knowledge source, and a refusal it must never break.



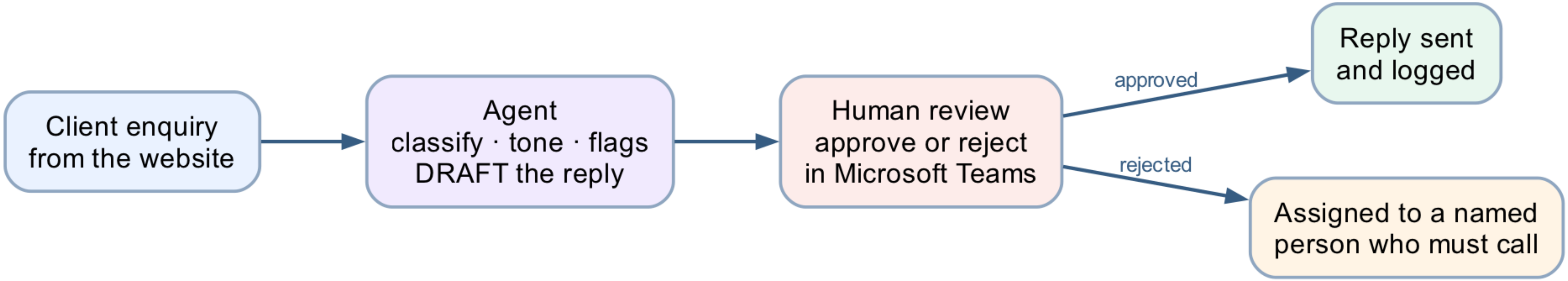
LAB 7

Lab 7 — HTTP and Chatbot

Investment advisor chatbot · Copilot Studio + SharePoint knowledge + static website · 40 min

Lab 8 — The Human Review Gate

Human in the loop — the AI drafts, and the workflow physically cannot proceed until a person approves.



LAB 8

Lab 8 — HTTP and Human Review

Client rapport assistant with a Teams approval gate · Copilot Studio + Human review + Excel Online + Outlook + Teams · 60 min

MODULE 5

Retrieval Augmented Generation

Why not just paste everything · ingestion and retrieval · chunking, embeddings, top_k · built-in versus external · probing for invention

Applied in Lab 9 and Lab 10

Why Retrieval, Rather Than a Bigger Prompt

You could paste all 20 brochures into the instruction. For 20 it works. Then the academy adds 40 more.

EVERYTHING IN THE PROMPT

The instruction exceeds what the model can read.
It starts ignoring the middle.
Every question costs the price of 60 brochures.
A fee change means editing the prompt.

RETRIEVE, THEN GENERATE

Find the two or three brochures that resemble the question, and give the model only those.
The agent never sees the other 17.
It answers from what matters.

Question → (vector) → **Embed the question** → (similarity) → **Search for the nearest documents** → (top_k) → **Paste them into the prompt** → (generate) → **Grounded answer**

RAG = retrieval + generation. The retrieval decides whether the generation can possibly be right.

The RAG Pipeline, and the Four Words That Matter

Two phases. Ingestion happens once; retrieval happens on every question.

INGESTION — once

Documents → (split) → Chunk → (model) → Embed → (index) → Store as vectors

RETRIEVAL — every question

Question → (same model) → Embed → (similarity) → Nearest top_k → (context) → Into the prompt → Answer

Chunk

How a document is split. Here: one brochure = one record.

Embedding

Text as a list of numbers. llama-text-embed-v2, 1024 dimensions.

Similarity

Nearness of two vectors — the machine's idea of "related".

top_k

How many documents come back. Here: 3.

Built-in Knowledge or Your Own Vector Store

	LAB 9 — built-in knowledge	LAB 10 — Pinecone
Nodes	3	5
Ingestion	Upload to SharePoint	A Python script, run once
Embedding model	Hidden	llama-text-embed-v2, 1024
Chunking	Hidden	One brochure = one record — your call
top_k	Hidden	3 — your call
Citation markers	Leak into the reply	None
A changed fee	Edit, wait for re-crawl	Edit, then re-ingest
When it answers badly	Rewrite the prompt and hope	Four levers to pull

Neither is the right answer. Which is right depends on whether the person maintaining it will ever need those levers.

Probing for Invention

A grounded agent still invents. You find out by asking for things that do not exist.

A course you do not run

"Do you have a course on cake sculpture?"

It must say it does not — not improvise a syllabus.

A fact in no brochure

"Is there parking at the east campus?"

It must say the brochures do not cover it.

A discount that does not exist

"What is the student discount?"

It must not invent a percentage to be helpful.

A CONFIDENT WRONG ANSWER RAISES NO ERROR

The run is green. The reply is fluent. Nothing in the run history is red. In every lab here the wrong answer looks exactly like a right one — which is why the test tables include the probes they do.

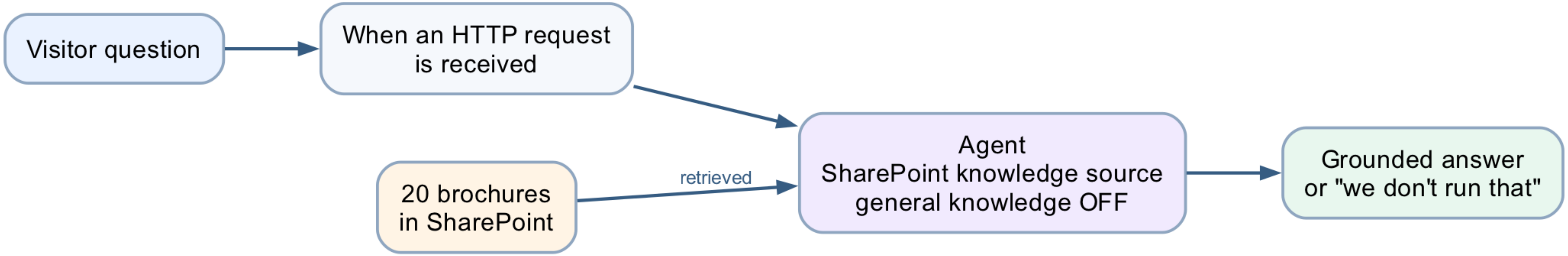
LAB 9

Lab 9 — RAG with Knowledge Base

Cook & Bake Academy, built-in knowledge · Copilot Studio Knowledge + SharePoint · 40 min

Lab 9 — RAG with Built-in Knowledge

RAG as a product setting — three nodes, no ingestion, and nothing you can inspect.



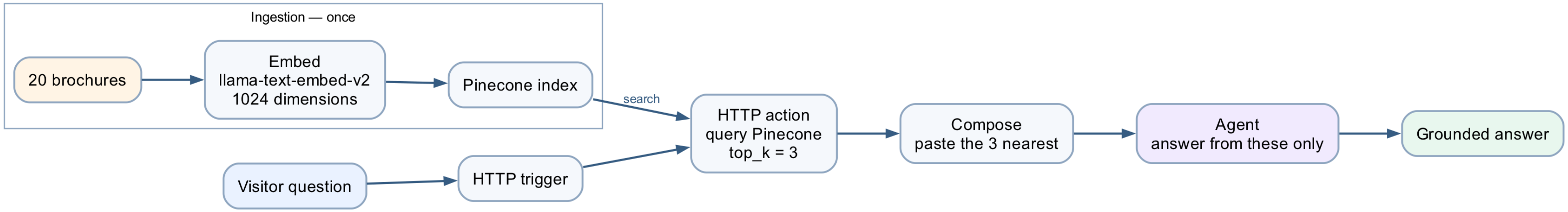
LAB 9

Lab 9 — RAG with Knowledge Base

Cook & Bake Academy, built-in knowledge · Copilot Studio Knowledge + SharePoint · 40 min

Lab 10 — RAG with Pinecone

What the convenience cost — chunking, embeddings, dimension and top_k, back in your hands.



LAB 10

Lab 10 — RAG with Pinecone

The same chatbot, with the levers back · Copilot Studio + Power Automate + Pinecone · 40 min

SYNTHESIS

Taking This Back to Work

Choosing the pattern · the pre-production checklist · the six sentences worth keeping

SYNTHESIS

Choosing the Right Pattern

Back at work, this is the decision you will actually make.

If the situation is…	…build this	Seen in
The rule is written down and never varies	**Workflow only**	**Labs 1–3**
The rule varies, but the outcome is high-stakes	**Agent decides, flow enforces, human approves**	**Labs 6, 8**
The input is language, the output is an answer	**Agent grounded in your documents**	**Labs 7, 9, 10**
Several audiences, different things they may know	**Connected agents, one per boundary**	**Lab 4**
The answer must be traceable to a source	**RAG, and turn general knowledge off**	**Labs 9, 10**

Ask who pays for the mistake. That answer sizes the agent's authority and places the human.

SYNTHESIS

Before Anything Reaches Production

1 Environment

Correct environment selected in BOTH products; Copilot Credits confirmed.

2 Data path

No model output writes to a system of record without a deterministic action in between.

3 Refusals tested

Every prohibition has a probe that tries to break it, and the probe is in the test set.

4 Human placed

Consequential paths end at a named person, not at a queue nobody watches.

5 Audit before the gate

The audit row is written before the approval, so rejected cases still leave a trace.

6 Sources closed

General knowledge off, "Search all websites" removed, knowledge status Ready.

A green run is not a correct run, and a fluent answer is not a true one. Verify both before anyone depends on it.

SYNTHESIS

Six Sentences Worth Keeping

1. A control the model cannot reach beats a rule you asked it to follow.
2. A reference to nothing resolves to empty, not to an error.
3. Commit the record of the obligation before you create the obligation.
4. The AI decides what to do; deterministic actions do what must always happen.
5. A pause that will still be paused tomorrow is a real gate.
6. Retrieval decides whether generation can possibly be right.

Six sentences. If you keep only these, you can rebuild the judgement behind every lab in this course.

Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

FINAL ASSESSMENT

Assessment & Funding

1 Written

1 hour · open book
Modules 1–5 and the lab concepts

2 Practical

1 hour · open book
Build and verify a working flow and agent

3 Evidence

Submit on the LMS
Competent result + attendance requirement

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

COURSE PORTAL

Courseware & Assessment on the LMS

- Download the slides and the Learner Guide from the LMS for the open-book assessment.
- Portal: https://lms-tms.tertiaryinfotech.com/
- Submit your Written Assessment and Practical Performance answers on the LMS.
- Complete the TRAQOM survey via the QR code on the LMS.

Assessment

- Written Assessment (SAQ) — 1 hour. Practical Performance (PP) — 1 hour.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at https://lms-tms.tertiaryinfotech.com/.

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Thank You

Now go automate something — questions welcome